MHF4U Advanced Functions - Unit 1 Practice Test, Version 2

Name: _____ Date: _____

Functions and Rates of Change | 75 minutes | 50 marks

Answer all questions. Show your work and give reasons where requested. Label graph features and use correct interval notation. Use the page 1 grids for Questions 1 and 2, choosing suitable scales. You may use a calculator. Formal differentiation rules are not required. The diagrams are original and are not drawn to scale unless stated.

1. Piecewise functions [9 marks]

The function P is defined by

```
            x + 3       if -4 <= x < 0
P(x) =      3 - x^2     if 0 <= x <= 2
            x - 1       if 2 < x <= 4
```

a. Sketch P on one set of axes. Label all open and closed endpoints. [K2 C2]

b. State its domain and range in interval notation. [K2]

c. State the intervals where P is increasing. [K1]

d. Would the inverse relation be a function? Explain using your graph. [T1 C1]

2. Absolute value graph [6 marks]

Sketch `f(x) = -2|x + 1| + 4`. Label the vertex and both x-intercepts. [K2 C1] State the domain and range in interval notation. [K2] Explain how the coefficient `-2` changes the graph of `y = |x|`. [T1]

3. Transforming a function [5 marks]

The point `(4, 2)` lies on `y = f(x)`, where `f(x) = sqrt(x)`. Define `g(x) = 3f(-2(x - 1)) - 1`.

a. Find the corresponding point on g, showing the input equation. [T2]

b. State the domain and range of g. [K2]

c. Describe the horizontal transformation. [C1]

4. Inverse relations [5 marks]

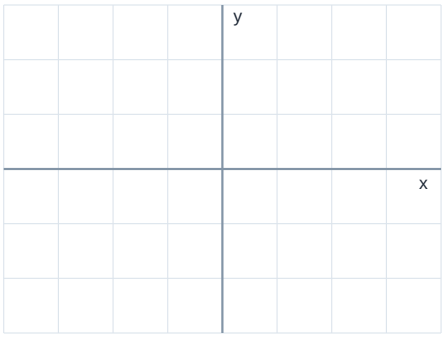
Let `q(x) = (x + 2)^2 - 1`, initially with domain all real numbers.

a. State the domain and range of its inverse relation. [K2]

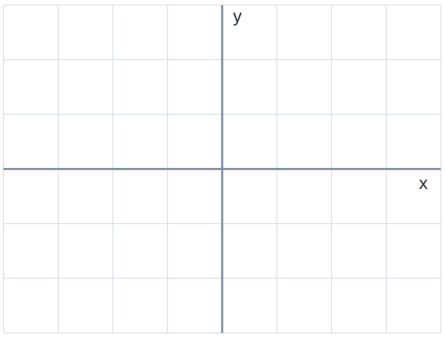
b. Explain why this inverse relation is not a function. [T1]

c. Restrict the domain of q so that its inverse becomes a function, then write that inverse function. [A2]

Q1 - piecewise graph

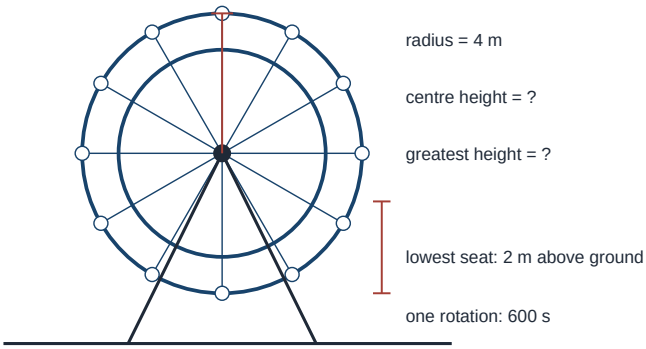


Q2 - absolute value graph



5. Observation wheel diagram [7 marks]

An observation wheel has radius 4 m. Its lowest seat is 2 m above the ground. It makes exactly one rotation in 600 s at constant speed. A rider begins at the lowest point at `t = 0`.



a. From the diagram, determine the centre height and greatest seat height. [K2]

b. State the domain and range of the rider's height function for this ride, in interval notation. [K2]

c. A student says the seat could reach 12 m because the wheel is 4 m above its centre. Explain the error. [T1]

d. Sketch a qualitative height-versus-time graph for one rotation. Label the height at `t = 0`, `t = 300`, and `t = 600`. [C2]

6. Average and instantaneous rates [8 marks]

During its first 6 seconds, a ball's height above the ground is modelled by `H(t) = -4t^2 + 24t + 1` metres.

a. Calculate the average rate of change of height over `[0, 2]`. [A2]

b. Estimate the instantaneous rate at `t = 2` using the centred interval `[1.9, 2.1]`. Show the two heights used. [A2 T1]

c. At what time does the ball reach its maximum modelled height, and what is that height? [K2]

d. Explain why your answers in a and b differ. [C1]

7. Graph properties and interval notation [5 marks]

Let `r(x) = x^3 - 4x`.

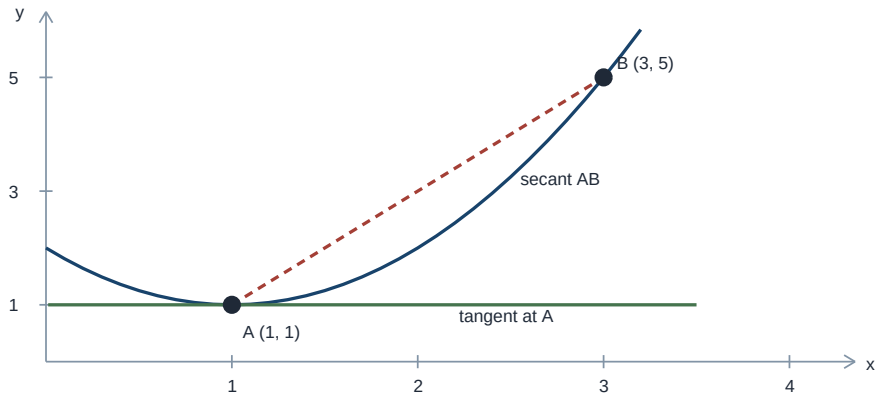
a. Show algebraically whether `r` is even, odd, or neither. [T2]

b. If `(3, 15)` lies on its graph, use symmetry to name one more point. [K1]

c. Write the solution to `|x - 1| <= 3` in interval notation. [K2]

8. Secant and tangent diagram [5 marks]

The graph shows `y = (x - 1)^2 + 1`, a secant through `A(1,1)` and `B(3,5)`, and the tangent at A.



a. Which line represents the average rate of change on `[1,3]`? Calculate its slope. [K1 A1]

b. Which line represents the instantaneous rate at `x = 1`? State its slope. [K1 A1]

c. Explain why the two rates differ even though both lines touch the graph at A. [T1]